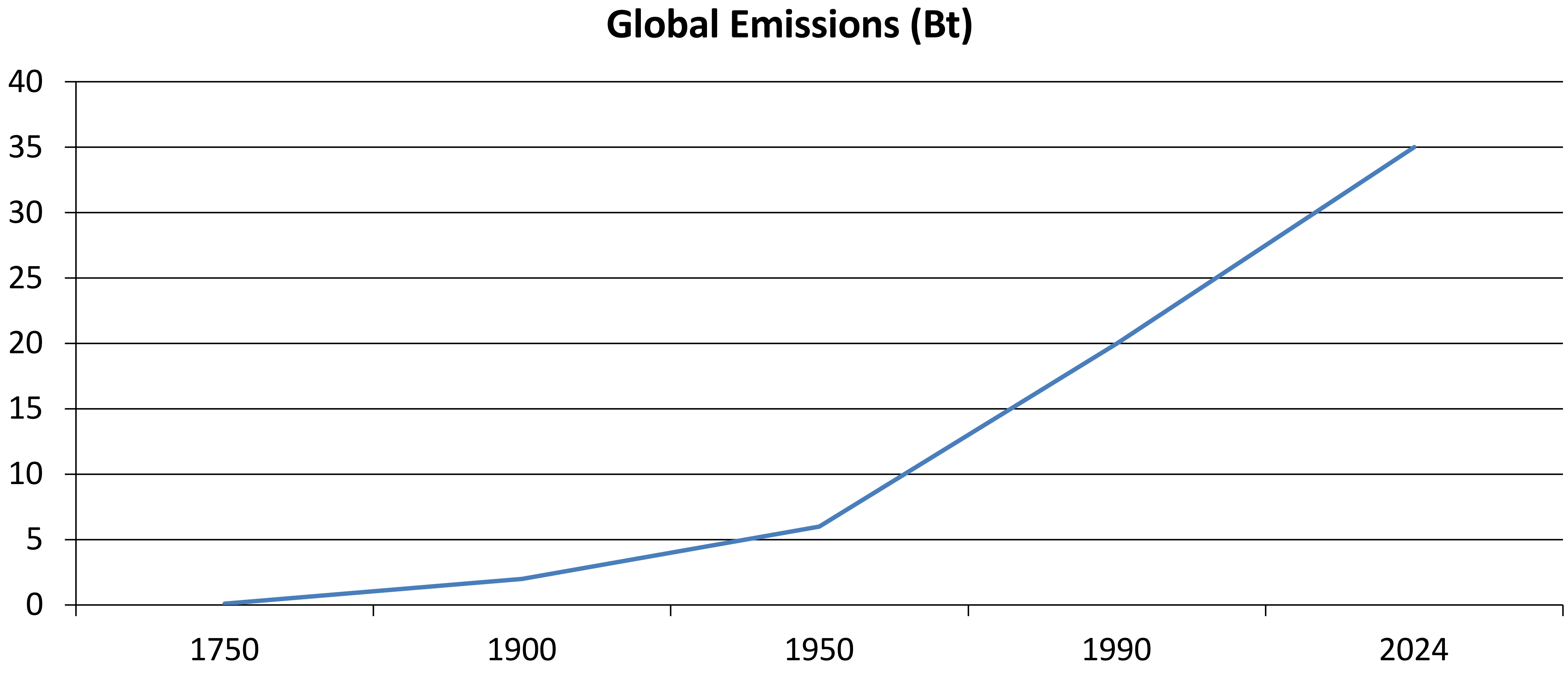
Global CO_2 Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

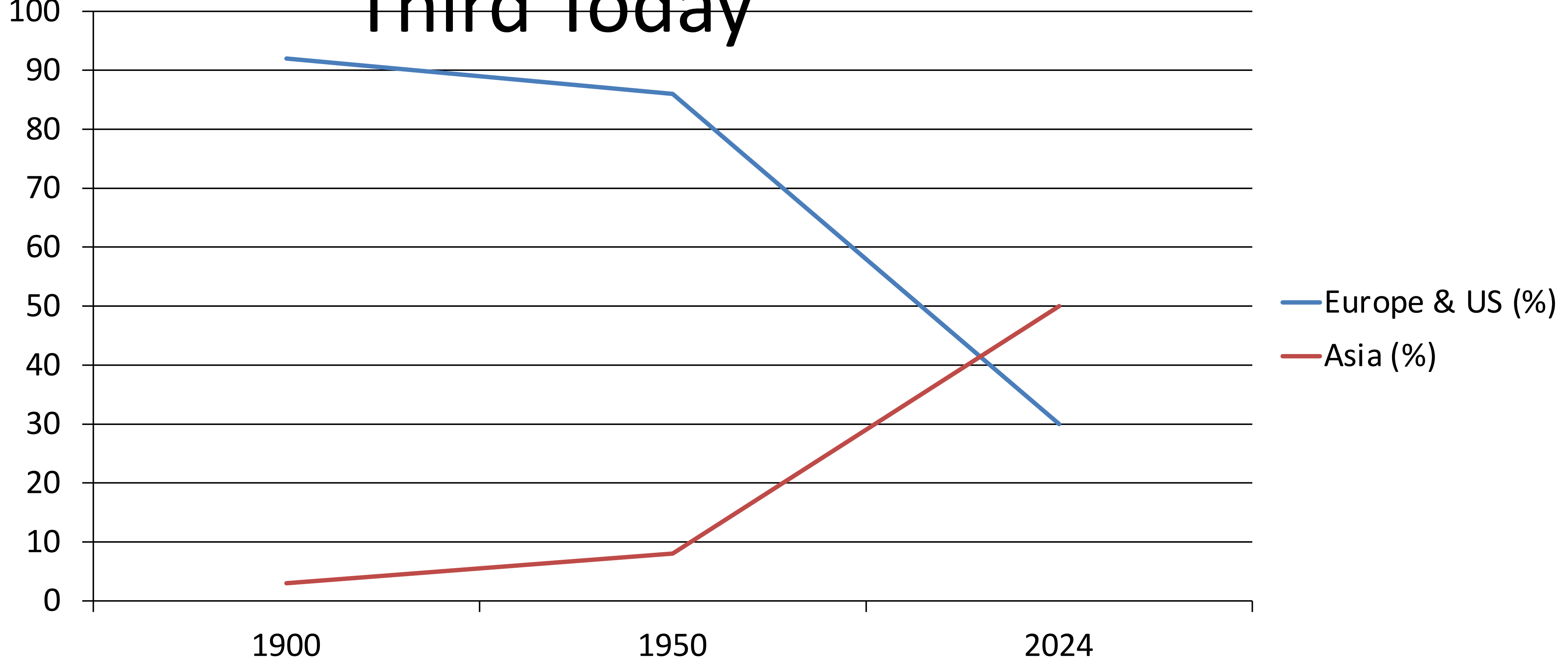
For policy analysts, climate negotiators, and sustainability researchers

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

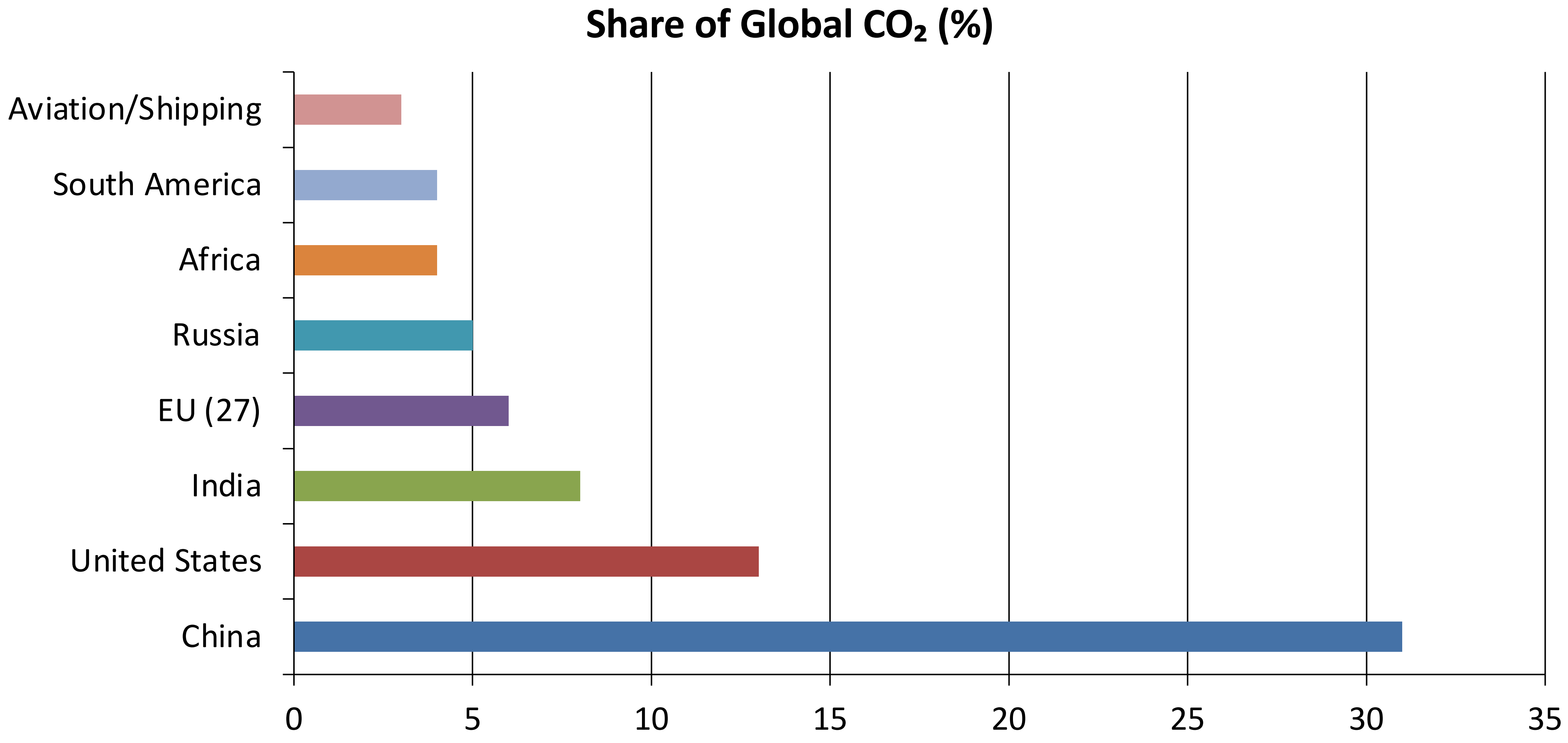


Inflection Points: Slow pre-industrial growth, post-1950 acceleration, recent slowing but no peak.

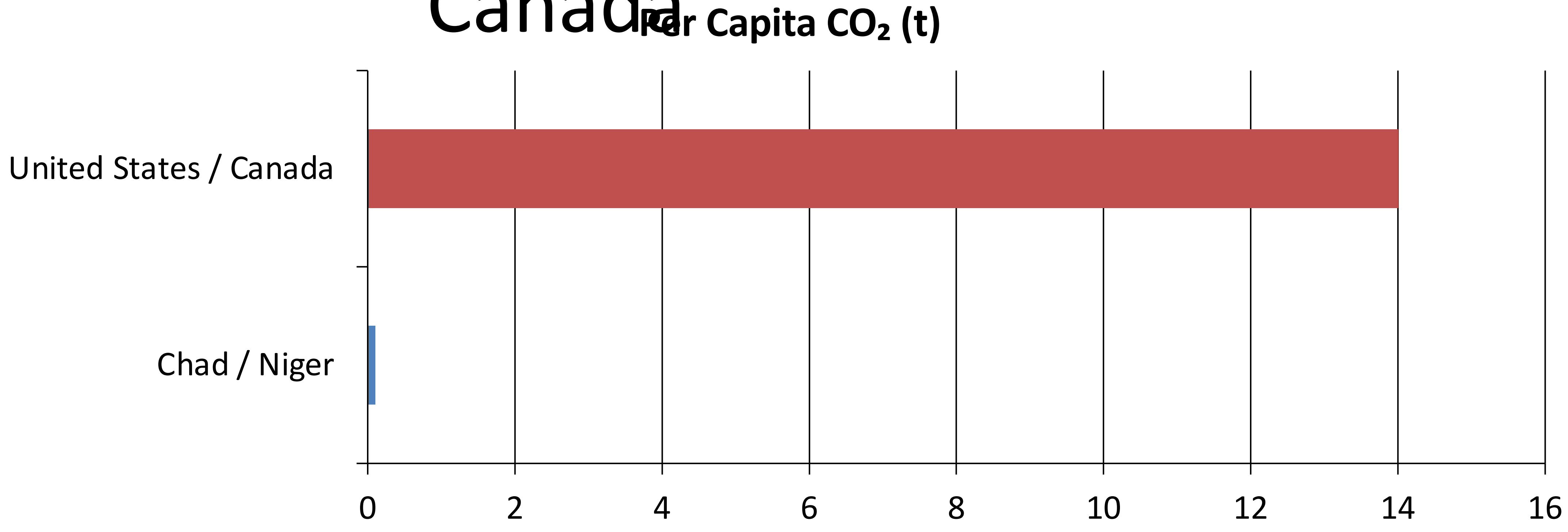
Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



China Now Emits More Than One-Quarter of Global CO_2

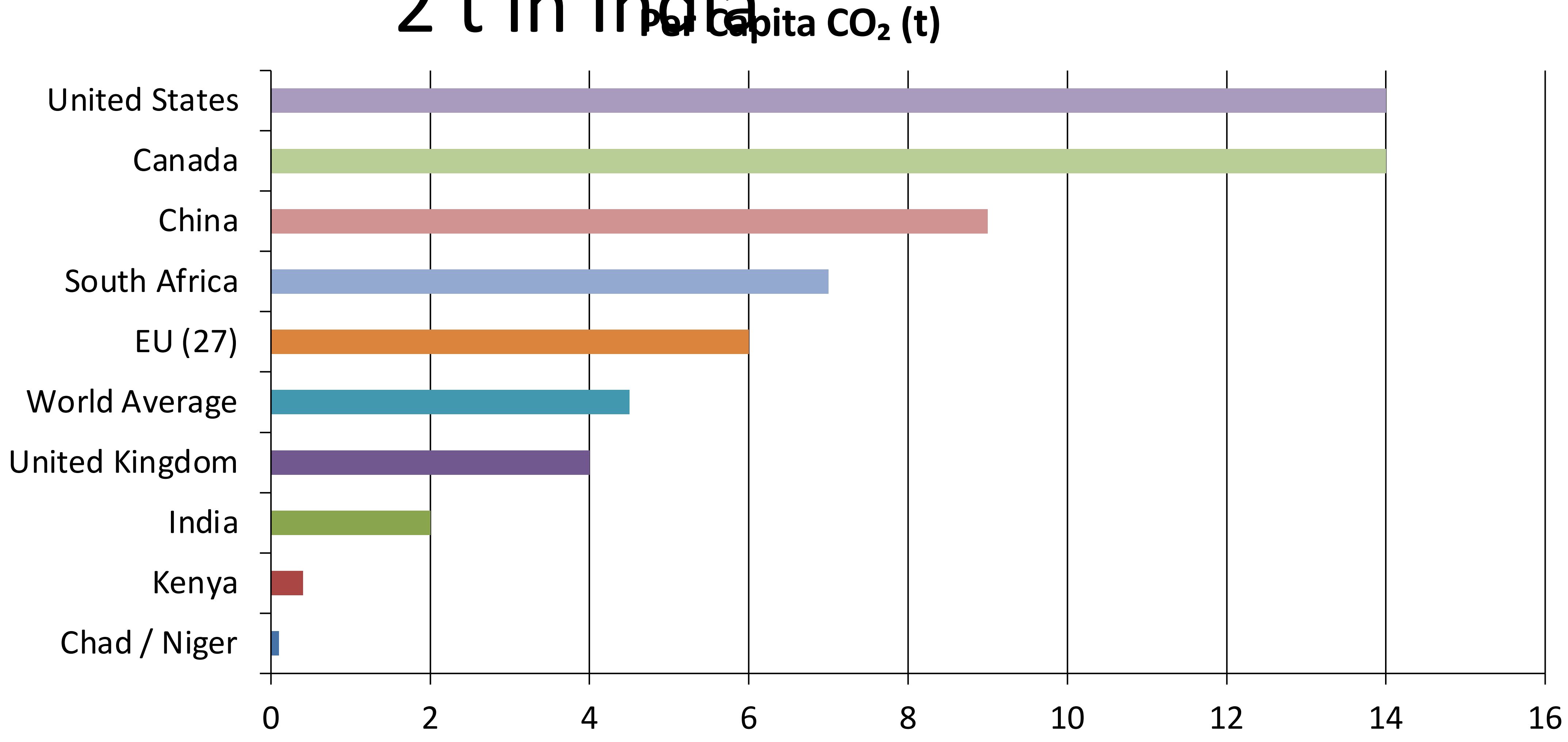


Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



A 150x gap exists between the poorest Sub-Saharan African nations and the US/Canada.
European countries like France and the UK are much lower than the US despite similar living standards.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

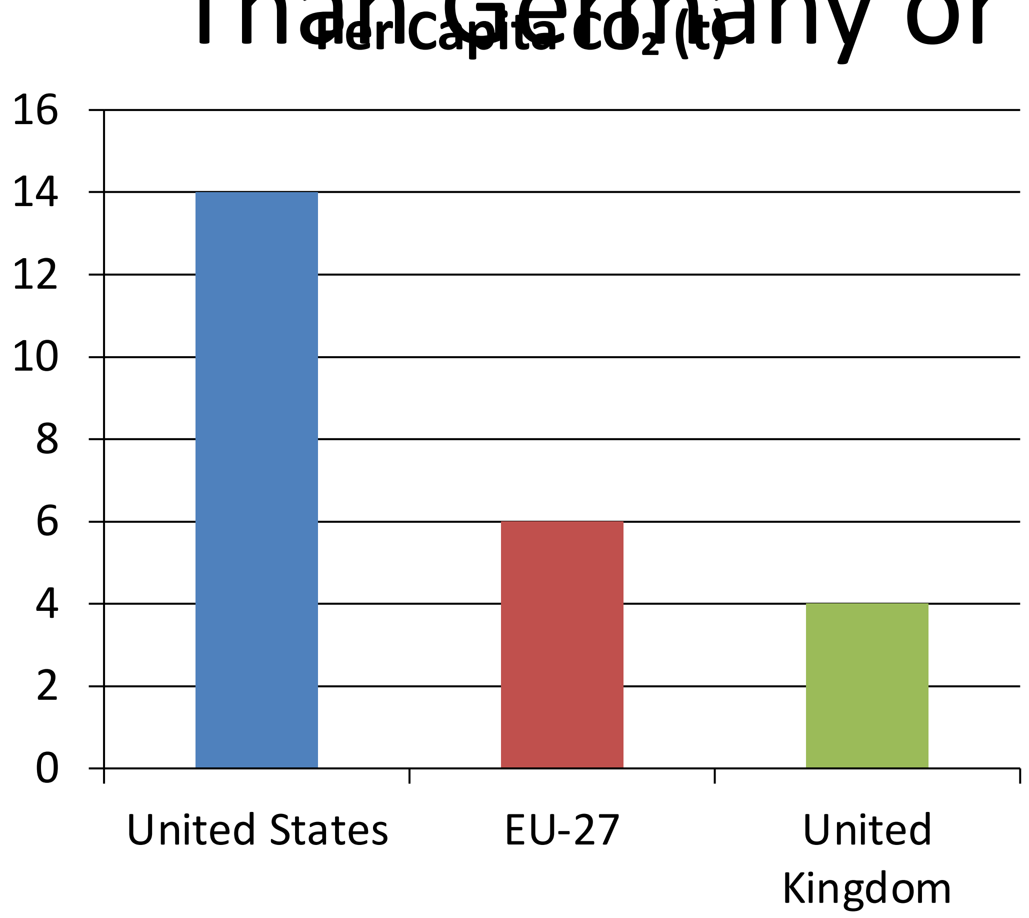


2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

[Placeholder: Diverging Bar Chart of Annual Percentage Change in CO_2 Emissions, 2024]

Declining Emissions (Blue)	Increasing Emissions (Red/Orange)
• United States • European Union (most) • United Kingdom	• Russia • Parts of Southeast Asia • Several African nations • Central Asia (e.g., Turkmenistan/Uzbekistan >10%)

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



Among wealthy nations with similar prosperity levels, per capita emissions vary significantly because of energy mix choices.
France, the UK, and Portugal have much lower per capita emissions than Germany, the Netherlands, or the US.
This is largely because a higher share of their electricity comes from nuclear and renewable sources rather than fossil fuels.

Key Takeaways: Responsibility Is Shared but Unequal

- Global emissions exceed 35 Bt/yr and haven't peaked, despite slowing growth.
- China is the largest annual emitter, but its per capita emissions remain below the US.
- Historical responsibility is dominated by Europe and the US.
- Massive per capita gaps of 150x persist between the richest and poorest nations.
- Energy policy and technology choices can decouple prosperity from emissions.